MATHCLEAR · VTU 2021 SCHEME

21MATCS41 Complete Revision Handbook

Mathematical Foundations for Computing, Probability and Statistics

All 30 practice parts with plain-English intuition, formula meanings, stepwise VTU-style working, mark allocation and final answers.

PREPARATION TARGET

60+ raw marks

Prepare above the minimum. No handbook can guarantee an examination result.

How to use this book so the answer stays in your head

Read: Read the plain-English idea once.

Cover: Hide the formal solution.

Recall: Say the formula and next step aloud.

Write: Reproduce the answer without looking.

Check: Compare every line and correct mistakes.

> **Important:** Passive reading alone cannot guarantee a pass. Writing each selected answer at least once is the reliable preparation method.

Source and verification policy

Primary authority: official VTU 2021 Scheme syllabus, official VTU model-paper structure and supplied VTU examination papers.

Secondary cross-check: Take It Easy Engineers SIMP/MQP listings are used only to compare importance—not to override the VTU syllabus or claim exact predictions.

Solutions: independently written in a VTU-style sequence; not an official VTU evaluation key.

Full forms and formula language

VTU Visvesvaraya Technological University	**SEE** Semester End Examination
CIE Continuous Internal Evaluation	**PMF** Probability Mass Function — probabilities for discrete values
PDF Probability Density Function — density for a continuous variable	**SD** Standard Deviation — typical spread around the mean
H_0 Null Hypothesis — the claim tested first	**H_1** Alternative Hypothesis — the competing claim
SE Standard Error — expected sample-to-sample variation	**df** Degrees of Freedom — independent information used by a test
χ^2 Chi-square — goodness-of-fit test statistic	**r / ρ** Correlation coefficient — strength and direction of association

How to read the common formulas

$z = (\bar{x} - \mu_0)/(\sigma/\sqrt{n})$: observed mean difference divided by its standard error.

$t = (\bar{x} - \mu_0)/(s/\sqrt{n})$: small-sample mean test when population SD is unknown.

$\chi^2 = \Sigma(O - E)^2/E$: add the squared observed–expected differences divided by expected counts.

$P(X=x)=e^{-\lambda}\lambda^x/x!$: Poisson probability for exactly x events when the mean is λ.

$\rho = 1 - 6\Sigma d^2/[n(n^2-1)]$: Spearman correlation from differences between two rankings.

21MATCS41

VTU-Aligned Practice Paper · 2021 Scheme

MATHEMATICAL FOUNDATIONS FOR COMPUTING, PROBABILITY AND STATISTICS

Time: 3 Hours **Max. Marks: 100**

Note: Answer any FIVE full questions, choosing ONE full question from each module. Use statistical tables where necessary.

MODULE – 1

Fundamentals of Logic

3/3 scanned papers: tautology, inference, equivalence and quantifiers

Q1

20 Marks

1(a) Define tautology. Show by truth table that $[(p\to q)\wedge(q\to r)]\to(p\to r)$ is a tautology. **[06 Marks]**

Tautology · 3/3 papers

FIRST UNDERSTAND THIS IN PLAIN ENGLISH

A tautology is a statement that never becomes false. Check all eight truth-value rows; the final column must contain only T.

VTU ANSWER ORDER

1. Given / definition
2. Formula / theorem
3. Substitution
4. Stepwise working
5. Boxed answer

VTU-STYLE STEPWISE WORKING	MARKS
STEP 1 **Define tautology** **Why?** First say what the word means. This gives the examiner the rule you will use next. A compound proposition true for every assignment of truth values.	1
STEP 2 **List eight combinations** **Why?** List every possible True/False case so no case is accidentally missed. $(p, q, r) = TTT, TTF, TFT, TFF, FTT, FTF, FFT, FFF$	1

STEP 3 **Compute p→q and q→r** Why? This line moves the solution one step closer to the answer. Copy the setup first, then simplify only one part at a time. An implication is false only when its first part is true and second part is false.	1
STEP 4 **Compute conjunction and p→r** Why? This line moves the solution one step closer to the answer. Copy the setup first, then simplify only one part at a time. Form the two columns needed by the outer implication.	1
STEP 5 **Compute final implication** Why? Write the cleaned-up result clearly and box it so the examiner can find it immediately. The final column is T in all eight rows.	1
STEP 6 **Conclude** Why? This line moves the solution one step closer to the answer. Copy the setup first, then simplify only one part at a time. $\boxed{[(p \to q) \wedge (q \to r)] \to (p \to r) \text{ is a tautology}}$	1
Total	6/6

FINAL ANSWER — WRITE THIS CLEARLY AND BOX IT

TAUTOLOGY

1(b) Test: If a program compiles, it executes. If it executes, its output can be tested. Its output cannot be tested. Therefore it does not compile. [07 Marks]

Rules of inference · 3/3 papers

FIRST UNDERSTAND THIS IN PLAIN ENGLISH

Give each sentence a letter. Chain the first two arrows, then use the negative ending to move backward.

VTU ANSWER ORDER

1. Given / definition	2. Formula / theorem
3. Substitution	4. Stepwise working
5. Boxed answer	

VTU-STYLE STEPWISE WORKING	MARKS
STEP 1 **Assign propositions** **Why?** Replace each long sentence with a short letter. This makes the logic easier to follow. $p = \text{compiles},\ q = \text{executes},\ r = \text{testable}$	1
STEP 2 **Translate premises** **Why?** Replace each long sentence with a short letter. This makes the logic easier to follow. $p \to q, \quad q \to r, \quad \neg r$	2
STEP 3 **Chain implications** **Why?** This line moves the solution one step closer to the answer. Copy the setup first, then simplify only one part at a time. $p \to r \quad (\text{hypothetical syllogism})$	1

STEP 4 **Apply modus tollens** **Why?** This line moves the solution one step closer to the answer. Copy the setup first, then simplify only one part at a time. $p \to r, \neg r \therefore \neg p$	2
STEP 5 **Conclusion** **Why?** State the decision in words and connect it back to what the question actually asked. $\boxed{\text{The argument is valid}}$	1
Total	7/7

FINAL ANSWER — WRITE THIS CLEARLY AND BOX IT

VALID;
$\neg p$

1(c) Prove directly: if n is odd, then n^2 is odd and n^2+1 is even. **[07 Marks]**

Direct proof · 2/3 papers + model paper

FIRST UNDERSTAND THIS IN PLAIN ENGLISH

Odd numbers always look like 2k+1. Square that form and rearrange it into "two times an integer plus one."

VTU ANSWER ORDER

1. Given / definition
2. Formula / theorem
3. Substitution
4. Stepwise working
5. Boxed answer

VTU-STYLE STEPWISE WORKING	MARKS
STEP 1 **Write odd form** **Why?** Rewrite the number using 2k for even or 2k+1 for odd, then rearrange it into the required form. $n = 2k + 1,\ k \in \mathbb{Z}$	1
STEP 2 **Square it** **Why?** This line moves the solution one step closer to the answer. Copy the setup first, then simplify only one part at a time. $n^2 = 4k^2 + 4k + 1 = 2(2k^2 + 2k) + 1$	2
STEP 3 **Conclude n^2 odd** **Why?** Rewrite the number using 2k for even or 2k+1 for odd, then rearrange it into the required form. It has the form 2m+1.	1
STEP 4 **Add one** **Why?** This line moves the solution one step closer to the answer. Copy the setup first, then simplify only one part at a time. $n^2 + 1 = 2(2k^2 + 2k + 1)$	2

STEP 5

Conclude

Why? This line moves the solution one step closer to the answer. Copy the setup first, then simplify only one part at a time.

$$\boxed{n^2 \text{ odd and } n^2 + 1 \text{ even}}$$

	1
Total	7/7

FINAL ANSWER — WRITE THIS CLEARLY AND BOX IT

N^2 ODD, $N^2 + 1$ EVEN

OR

Q2

20 Marks

2(a) Show by truth table that [(p→q)∧p]→q is a tautology. [06 Marks]

Tautology by truth table · 3/3 papers

FIRST UNDERSTAND THIS IN PLAIN ENGLISH

This is modus ponens written as one formula. Check all four p,q rows; the final column must never become false.

VTU ANSWER ORDER

1. Given / definition
2. Formula / theorem
3. Substitution
4. Stepwise working
5. Boxed answer

VTU-STYLE STEPWISE WORKING	MARKS

STEP 1 **Write four combinations** **Why?** List every possible True/False case so no case is accidentally missed. $$(p, q) = TT, TF, FT, FF$$	1
STEP 2 **Compute p→q** **Why?** This line moves the solution one step closer to the answer. Copy the setup first, then simplify only one part at a time. It is false only for p=T and q=F.	1
STEP 3 **Compute the conjunction** **Why?** This line moves the solution one step closer to the answer. Copy the setup first, then simplify only one part at a time. Find (p→q)∧p for each row.	1
STEP 4 **Compute final implication** **Why?** Write the cleaned-up result clearly and box it so the examiner can find it immediately. The final values are T,T,T,T.	2
STEP 5 **Conclude** **Why?** This line moves the solution one step closer to the answer. Copy the setup first, then simplify only one part at a time. $$\boxed{[(p \to q) \wedge p] \to q \text{ is a tautology}}$$	1
Total	6/6

FINAL ANSWER — WRITE THIS CLEARLY AND BOX IT

TAUTOLOGY

2(b) Over integers, find truth values: (i) $\forall x, x^2 \geq 0$ (ii) $\exists x, x^2=2$ (iii) $\forall x$, even(x)$\to$even(x^2) (iv) $\exists x, x<0$ and $x^2=9$. [07 Marks]

Quantifiers · 3/3 papers

FIRST UNDERSTAND THIS IN PLAIN ENGLISH

"For all" needs every integer to work. "There exists" needs only one example. A single counterexample defeats "for all."

VTU ANSWER ORDER

1. Given / definition
2. Formula / theorem
3. Substitution
4. Stepwise working
5. Boxed answer

VTU-STYLE STEPWISE WORKING	MARKS
STEP 1 **Statement (i)** **Why?** This line moves the solution one step closer to the answer. Copy the setup first, then simplify only one part at a time. True: every integer square is nonnegative.	1
STEP 2 **Statement (ii)** **Why?** This line moves the solution one step closer to the answer. Copy the setup first, then simplify only one part at a time. False: no integer has square 2.	2

STEP 3 **Statement (iii)** Why? This line moves the solution one step closer to the answer. Copy the setup first, then simplify only one part at a time. $x = 2k \Rightarrow x^2 = 2(2k^2) \Rightarrow \text{ true}$	2
STEP 4 **Statement (iv)** Why? This line moves the solution one step closer to the answer. Copy the setup first, then simplify only one part at a time. $x = -3 \text{ is a witness, so true}$	1
STEP 5 **Final list** Why? Write the cleaned-up result clearly and box it so the examiner can find it immediately. $\boxed{T, F, T, T}$	1
Total	7/7

FINAL ANSWER — WRITE THIS CLEARLY AND BOX IT

T
,
F
,
T
,
T

2(c) Write symbolically and negate: “Every integer divisible by 4 is even.” State whether it is true. [07 Marks]

Symbolic form and negation · 3/3 papers

FIRST UNDERSTAND THIS IN PLAIN ENGLISH

Negating “every” changes it to “there is at least one,” and the arrow becomes “first condition true but second false.”

VTU ANSWER ORDER

1. Given / definition
2. Formula / theorem
3. Substitution
4. Stepwise working
5. Boxed answer

VTU-STYLE STEPWISE WORKING	MARKS
STEP 1 **Declare universe** **Why?** This line moves the solution one step closer to the answer. Copy the setup first, then simplify only one part at a time. $n \in \mathbb{Z}$	1
STEP 2 **Symbolic statement** **Why?** This line moves the solution one step closer to the answer. Copy the setup first, then simplify only one part at a time. $\forall n,\ 4 \mid n \to 2 \mid n$	2
STEP 3 **Negation** **Why?** This line moves the solution one step closer to the answer. Copy the setup first, then simplify only one part at a time. $\exists n,\ 4 \mid n \land 2 \nmid n$	2

STEP 4

Verify

Why? Put the answer back into the original rule to make sure it really works.

1

$$n = 4k = 2(2k) \Rightarrow 2 \mid n$$

STEP 5

Conclusion

Why? State the decision in words and connect it back to what the question actually asked.

1

$$\boxed{\text{Original statement is true}}$$

Total 7/7

FINAL ANSWER — WRITE THIS CLEARLY AND BOX IT

∀
N
,
4 N 2
| →| N
IS TRUE

MODULE - 2

Relations, Functions and Graph Theory

Every option combines a function, relation/poset and graph problem

Q3

20 Marks

3(a) If |A|=4 and |B|=6, find all, one-one and onto functions A→B and B→A. [06 Marks]

Counting functions · 6/6 option sides

FIRST UNDERSTAND THIS IN PLAIN ENGLISH

For each input choose an output. One-one means no output repeats. Onto means every output is hit at least once.

VTU ANSWER ORDER

1. Given / definition
2. Formula / theorem
3. Substitution
4. Stepwise working
5. Boxed answer

VTU-STYLE STEPWISE WORKING	MARKS
STEP 1 **All A→B functions** **Why?** This line moves the solution one step closer to the answer. Copy the setup first, then simplify only one part at a time. $6^4 = 1296$	1
STEP 2 **One-one A→B** **Why?** This line moves the solution one step closer to the answer. Copy the setup first, then simplify only one part at a time. ${}^6P_4 = 360$	1

STEP 3 **Onto A→B** **Why?** This line moves the solution one step closer to the answer. Copy the setup first, then simplify only one part at a time. $0 \quad (4 < 6)$	1
STEP 4 **All B→A functions** **Why?** This line moves the solution one step closer to the answer. Copy the setup first, then simplify only one part at a time. $4^6 = 4096$	1
STEP 5 **One-one B→A** **Why?** This line moves the solution one step closer to the answer. Copy the setup first, then simplify only one part at a time. $0 \quad (6 > 4)$	1
STEP 6 **Onto B→A by inclusion-exclusion** **Why?** This line moves the solution one step closer to the answer. Copy the setup first, then simplify only one part at a time. $4^6 - \binom{4}{1}3^6 + \binom{4}{2}2^6 - \binom{4}{3}1^6 = 1560$	1
Total	6/6

FINAL ANSWER — WRITE THIS CLEARLY AND BOX IT

(
1296
,
360
,
0
)
;
(
4096
,
0
$A\ BB\ A,$

$\to: \to\ :$ 1560
)

3(b) On A={1,2,3,6}, define xRy iff x divides y. Prove it is a poset and draw the Hasse diagram. [07 Marks]

Poset and Hasse diagram · 2/3 papers + models

FIRST UNDERSTAND THIS IN PLAIN ENGLISH

Divisibility creates levels: 1 is below everything, 6 is above 2 and 3. Remove loop and shortcut arrows to get the Hasse diagram.

VTU ANSWER ORDER

1. Given / definition	2. Formula / theorem
3. Substitution	4. Stepwise working
5. Boxed answer	

VTU-STYLE STEPWISE WORKING	MARKS
STEP 1 **Write relation** Why? This line moves the solution one step closer to the answer. Copy the setup first, then simplify only one part at a time. $R = \{(1,1),(1,2),(1,3),(1,6),(2,2),(2,6),(3,3),(3,6),(6,6)\}$	2

STEP 2 **Reflexive** **Why?** A partial order needs three checks: every item relates to itself, two-way related items must be equal, and relation chains must continue. $x \mid x$	1
STEP 3 **Antisymmetric** **Why?** A partial order needs three checks: every item relates to itself, two-way related items must be equal, and relation chains must continue. For positive integers, x|y and y|x imply x=y.	1
STEP 4 **Transitive** **Why?** A partial order needs three checks: every item relates to itself, two-way related items must be equal, and relation chains must continue. x|y and y|z imply x|z.	1
STEP 5 **Cover relations** **Why?** Remove loops, arrowheads, and shortcut edges. Keep only immediate neighbours, then place larger elements above smaller ones. $1 \prec 2, 1 \prec 3, 2 \prec 6, 3 \prec 6$	1
STEP 6 **Hasse levels** **Why?** Remove loops, arrowheads, and shortcut edges. Keep only immediate neighbours, then place larger elements above smaller ones. $\boxed{\text{1 below 2, 3 below 6}}$	1

Total	7/7

FINAL ANSWER — WRITE THIS CLEARLY AND BOX IT

$\mathbf{2\ 3\ 6}$

$\mathbf{,\ ,\ ,}$

$\mathbf{1\ 1\ 2\ 3}$

$\prec\prec\prec\prec$

$\mathbf{6}$

3(c) Prove that the number of odd-degree vertices in a finite undirected graph is even. **[07 Marks]**

Degree theorem · Graph theory in 6/6 option sides

FIRST UNDERSTAND THIS IN PLAIN ENGLISH

Every edge gives two degree-counts, so the total degree is even. Odd numbers can add to an even number only when there are an even number of them.

VTU ANSWER ORDER

1. Given / definition
2. Formula / theorem
3. Substitution
4. Stepwise working
5. Boxed answer

VTU-STYLE STEPWISE WORKING	MARKS
STEP 1 **Handshaking lemma** **Why?** This line moves the solution one step closer to the answer. Copy the setup first, then simplify only one part at a time. $\sum_{v \in V} \deg(v) = 2\|E\| \text{ is even}$	2

STEP 2 **Separate parity groups** **Why?** Rewrite the number using 2k for even or 2k+1 for odd, then rearrange it into the required form. $$\sum_V \deg v = \sum_{V_e} \deg v + \sum_{V_o} \deg v$$	1
STEP 3 **Even-degree sum** **Why?** Check whether the graph mirrors itself. Even symmetry removes sine terms; odd symmetry removes cosine terms and saves work. The first sum is even.	1
STEP 4 **Odd-degree sum** **Why?** Check whether the graph mirrors itself. Even symmetry removes sine terms; odd symmetry removes cosine terms and saves work. The second sum must be even; a sum of odd integers is even only when their count is even.	2
STEP 5 **Conclusion** **Why?** State the decision in words and connect it back to what the question actually asked. $$\boxed{|V_o| \text{ is even}}$$	1
Total	7/7

FINAL ANSWER — WRITE THIS CLEARLY AND BOX IT

|
V
O
|
IS EVEN

OR

Q4

20 Marks

4(a) For f(x)=3x−4, find f^{-1} and verify both compositions. [06 Marks]

Inverse function · Exact family repeated

FIRST UNDERSTAND THIS IN PLAIN ENGLISH

Call the output y, solve backward for x, then swap the letter. Composition checks that going forward then backward returns the start.

VTU ANSWER ORDER

1. Given / definition
2. Formula / theorem
3. Substitution
4. Stepwise working
5. Boxed answer

VTU-STYLE STEPWISE WORKING	MARKS
STEP 1 **Set y=f(x)** **Why?** This line moves the solution one step closer to the answer. Copy the setup first, then simplify only one part at a time. $y = 3x - 4$	1

STEP 2 **Solve for x** **Why?** Now do the arithmetic or algebra one small operation at a time, keeping the previous line visible. $x = (y + 4)/3$	1
STEP 3 **State inverse** **Why?** This line moves the solution one step closer to the answer. Copy the setup first, then simplify only one part at a time. $f^{-1}(x) = (x + 4)/3$	1
STEP 4 **Verify f∘f⁻¹** **Why?** Put the answer back into the original rule to make sure it really works. $f((x + 4)/3) = x$	1.5
STEP 5 **Verify f⁻¹∘f** **Why?** Put the answer back into the original rule to make sure it really works. $f^{-1}(3x - 4) = x$	1.5
Total	6/6

FINAL ANSWER — WRITE THIS CLEARLY AND BOX IT

F
1
$($
X X
$)$
$= \frac{\pm}{3}$
4

4(b) On A={1,2,3}, let aRb when a and b have the same parity. Write R and its matrix, prove equivalence and give the partition. **[07 Marks]**

Equivalence relation · 2/3 papers

FIRST UNDERSTAND THIS IN PLAIN ENGLISH

Numbers are friends when both are odd or both are even. Friend groups become equivalence classes.

VTU ANSWER ORDER

1. Given / definition	2. Formula / theorem
3. Substitution	4. Stepwise working
5. Boxed answer	

VTU-STYLE STEPWISE WORKING	MARKS
STEP 1 **Ordered pairs** **Why?** Write the standard rule first. It tells the examiner exactly which method you are using. $R = \{(1,1),(1,3),(2,2),(3,1),(3,3)\}$	2

STEP 2 **Matrix** Why? Use 1 when the relation exists and 0 when it does not. Read rows as starting elements and columns as ending elements. $M_R = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}$	1
STEP 3 **Reflexive** Why? A partial order needs three checks: every item relates to itself, two-way related items must be equal, and relation chains must continue. Every number has its own parity.	1
STEP 4 **Symmetric** Why? Rewrite the number using 2k for even or 2k+1 for odd, then rearrange it into the required form. Same parity works in both directions.	1
STEP 5 **Transitive** Why? A partial order needs three checks: every item relates to itself, two-way related items must be equal, and relation chains must continue. If a matches b and b matches c, a matches c.	1
STEP 6 **Partition** Why? This line moves the solution one step closer to the answer. Copy the setup first, then simplify only one part at a time. $\boxed{\{\{1,3\},\{2\}\}}$	1

Total	7/7

FINAL ANSWER — WRITE THIS CLEARLY AND BOX IT

{{
1
,
3
}
,
{
2
}}

4(c) For V={a,b,c,d}, E={ab,bc,cd,da,ac}, find degrees and determine whether an Euler trail/circuit exists. [07 Marks]

Euler trail · Graph problem in 6/6 option sides

FIRST UNDERSTAND THIS IN PLAIN ENGLISH

Count edges touching each vertex. Exactly two odd vertices means an Euler trail starts at one and ends at the other.

VTU ANSWER ORDER

1. Given / definition	2. Formula / theorem
3. Substitution	4. Stepwise working
5. Boxed answer	

VTU-STYLE STEPWISE WORKING	MARKS
STEP 1 **Find degrees** **Why?** Count the edges touching each vertex, then use the odd-degree rule to decide the path type. $\deg a = 3, \deg b = 2, \deg c = 3, \deg d = 2$	2

STEP 2

Odd vertices

> **Why?** Rewrite the number using 2k for even or 2k+1 for odd, then rearrange it into the required form.

$$a, c$$

1

STEP 3

Apply criterion

> **Why?** Write the standard rule first. It tells the examiner exactly which method you are using.

Exactly two odd vertices gives an Euler trail but no Euler circuit.

2

STEP 4

Give trail

> **Why?** Count the edges touching each vertex, then use the odd-degree rule to decide the path type.

$$a - b - c - d - a - c$$

1

STEP 5

Conclusion

> **Why?** State the decision in words and connect it back to what the question actually asked.

$$\boxed{\text{Euler trail exists; circuit does not}}$$

1

Total 7/7

FINAL ANSWER — WRITE THIS CLEARLY AND BOX IT

$ABCDA$

$-----C$

IS AN EULER TRAIL

MODULE - 3

Statistical Methods

3/3: rank/correlation, regression and least-squares curve fitting

Q5

20 Marks

5(a) Two judges rank five contestants as (1,2,3,4,5) and (2,1,3,5,4). Find Spearman's rank correlation. [06 Marks]

Spearman rank correlation · 3/3 papers

FIRST UNDERSTAND THIS IN PLAIN ENGLISH

Subtract the two ranks for each person, square the differences, add them, then use one formula.

VTU ANSWER ORDER

1. Given / definition
2. Formula / theorem
3. Substitution
4. Stepwise working
5. Boxed answer

VTU-STYLE STEPWISE WORKING	MARKS
STEP 1 **Differences** **Why?** Compare the two rankings item by item. Squaring removes minus signs and makes larger disagreements count more. $d = (-1, 1, 0, -1, 1)$	1
STEP 2 **Squares** **Why?** Compare the two rankings item by item. Squaring removes minus signs and makes larger disagreements count more. $d^2 = (1, 1, 0, 1, 1)$	1

STEP 3 **Total** **Why?** This line moves the solution one step closer to the answer. Copy the setup first, then simplify only one part at a time. $\sum d^2 = 4,\ n = 5$	1
STEP 4 **Formula** **Why?** Write the standard rule first. It tells the examiner exactly which method you are using. $\rho = 1 - \frac{6\sum d^2}{n(n^2-1)}$	1
STEP 5 **Substitute** **Why?** Put the known numbers or conditions into the formula carefully before simplifying. $\rho = 1 - 24/120$	1
STEP 6 **Answer** **Why?** Write the cleaned-up result clearly and box it so the examiner can find it immediately. $\boxed{\rho = 0.8}$	1
Total	6/6

FINAL ANSWER — WRITE THIS CLEARLY AND BOX IT

ρ

$=$

0.8

5(b) Fit y=a+bx+cx² to x=−2,−1,0,1,2 and y=9,4,1,0,1. Estimate y at x=3. **[07 Marks]**

Parabola fitting · 3/3 papers

FIRST UNDERSTAND THIS IN PLAIN ENGLISH

Build the three normal equations from column totals. Symmetric x-values make many odd-power totals zero.

VTU ANSWER ORDER

1. Given / definition
2. Formula / theorem
3. Substitution
4. Stepwise working
5. Boxed answer

VTU-STYLE STEPWISE WORKING	MARKS
STEP 1 **Compute totals** **Why?** Build the small table totals needed by the least-squares equations, then solve those equations for the curve constants. $\sum x = \sum x^3 = 0, \sum x^2 = 10, \sum x^4 = 34, \sum y = 15, \sum xy = -20,$	2
STEP 2 **Normal equations** **Why?** Build the small table totals needed by the least-squares equations, then solve those equations for the curve constants. $15 = 5a + 10c, \quad -20 = 10b, \quad 44 = 10a + 34c$	2

STEP 3

Solve coefficients

Why? Solve the small simultaneous equations to find the unknown numbers in the fitted curve. Substitute one result into another equation and check each value.

$$a = 1,\ b = -2,\ c = 1$$

2

STEP 4

Estimate

Why? Put the requested x-value into the fitted equation. This uses the model you just found to predict the missing y-value.

$$\boxed{y = (x-1)^2;\ y(3) = 4}$$

1

Total 7/7

FINAL ANSWER — WRITE THIS CLEARLY AND BOX IT

X
2
,
Y
(
$2\ 3$
$Y 1\ X)$
$= - + =$
4

5(c) Regression equations are x=0.5y+1 and y=0.8x+2. Find x,̄ ȳ and r. **[07 Marks]**

Regression equations · 3/3 papers

FIRST UNDERSTAND THIS IN PLAIN ENGLISH

Both regression lines cross at the two means. Solve them together, then multiply the two regression coefficients and take the square root.

VTU ANSWER ORDER

1. Given / definition
2. Formula / theorem

3. Substitution	4. Stepwise working
5. Boxed answer	

VTU-STYLE STEPWISE WORKING	MARKS
STEP 1 **Intersection principle** **Why?** Both regression lines pass through the point made by the two means. Solve the lines together to find that point. Regression lines meet at (x,y̅).	1
STEP 2 **Solve x̄** **Why?** Now do the arithmetic or algebra one small operation at a time, keeping the previous line visible. $x = 0.5(0.8x + 2) + 1 \Rightarrow \bar{x} = 10/3$	2
STEP 3 **Find ȳ** **Why?** This line moves the solution one step closer to the answer. Copy the setup first, then simplify only one part at a time. $\bar{y} = 0.8(10/3) + 2 = 14/3$	1
STEP 4 **Coefficients** **Why?** This coefficient measures how much of one sine or cosine wave is hidden inside the function. Substitute the function and integrate over the stated interval. $b_{xy} = 0.5,\ b_{yx} = 0.8$	1

STEP 5 **Correlation** **Why?** Compare how the two variables move together. Positive means they usually rise together; negative means one falls as the other rises. $\boxed{r = +\sqrt{0.4} = 0.6325}$	2
Total	7/7

FINAL ANSWER — WRITE THIS CLEARLY AND BOX IT

10/314/3
, ,
XY R
$==$ $=$
0.6325

OR

Q6

20 Marks

6(a) Find Karl Pearson's r for x=(1,2,3,4,5), y=(2,4,5,4,5). [06 Marks]

Karl Pearson correlation · 3/3 papers

FIRST UNDERSTAND THIS IN PLAIN ENGLISH

Measure how x and y move away from their averages together. Same-direction movement makes the product positive.

VTU ANSWER ORDER

1. Given / definition
2. Formula / theorem
3. Substitution
4. Stepwise working
5. Boxed answer

VTU-STYLE STEPWISE WORKING	MARKS

STEP 1 **Means** **Why?** Multiply each value by its probability and add. This finds the balance point of the distribution. $\bar{x} = 3, \bar{y} = 4$	1
STEP 2 **Deviation totals** **Why?** Build the small table totals needed by the least-squares equations, then solve those equations for the curve constants. $\sum dx\, dy = 6, \sum dx^2 = 10, \sum dy^2 = 6$	2
STEP 3 **Formula** **Why?** Write the standard rule first. It tells the examiner exactly which method you are using. $r = \dfrac{\sum dxdy}{\sqrt{\sum dx^2 \sum dy^2}}$	1
STEP 4 **Substitute** **Why?** Put the known numbers or conditions into the formula carefully before simplifying. $r = 6/\sqrt{60}$	1
STEP 5 **Answer** **Why?** Write the cleaned-up result clearly and box it so the examiner can find it immediately. $\boxed{r = 0.7746}$	1

Total	6/6

FINAL ANSWER — WRITE THIS CLEARLY AND BOX IT

$R = 0.7746$

6(b) Fit y=a+bx to x=(0,1,2,3), y=(1,3,5,7) and estimate y at x=5. **[07 Marks]**

Straight-line fitting · 3/3 papers

FIRST UNDERSTAND THIS IN PLAIN ENGLISH

Least squares chooses the line with the smallest total squared misses. Here the points already lie exactly on one line.

VTU ANSWER ORDER

1. Given / definition
2. Formula / theorem
3. Substitution
4. Stepwise working
5. Boxed answer

VTU-STYLE STEPWISE WORKING	MARKS
STEP 1 **Normal equations** **Why?** Build the small table totals needed by the least-squares equations, then solve those equations for the curve constants. $\sum y = na + b\sum x, \quad \sum xy = a\sum x + b\sum x^2$	2

STEP 2 **Totals** **Why?** Build the small table totals needed by the least-squares equations, then solve those equations for the curve constants. $$n = 4, \sum x = 6, \sum y = 16, \sum x^2 = 14, \sum xy = 34$$	1
STEP 3 **Substitute** **Why?** Put the known numbers or conditions into the formula carefully before simplifying. $$4a + 6b = 16, \quad 6a + 14b = 34$$	1
STEP 4 **Solve** **Why?** Now do the arithmetic or algebra one small operation at a time, keeping the previous line visible. $$a = 1, b = 2$$	2
STEP 5 **Estimate** **Why?** Put the requested x-value into the fitted equation. This uses the model you just found to predict the missing y-value. $$\boxed{y = 1 + 2x;\ y(5) = 11}$$	1
Total	7/7

FINAL ANSWER — WRITE THIS CLEARLY AND BOX IT

2
X
$,$
Y
$($
5
Y_1 $)$
$=+=$
11

6(c) Fit $y=ax^b$ to x=(1,2,4,8), y=(3,12,48,192). **[07 Marks]**

Power curve · 2/3 papers + model

FIRST UNDERSTAND THIS IN PLAIN ENGLISH

Take logarithms so powers become multiplication. Here doubling x multiplies y by four, so the power must be two.

VTU ANSWER ORDER

1. Given / definition
2. Formula / theorem
3. Substitution
4. Stepwise working
5. Boxed answer

VTU-STYLE STEPWISE WORKING	MARKS
STEP 1 **Linearize** **Why?** Taking logarithms turns a power curve into a straight-line equation, which is much easier to fit. $\log y = \log a + b \log x$	2

STEP 2 **Compare ratios** **Why?** Compare the calculated test value with the allowed boundary. Crossing the boundary means the difference is statistically significant. $x \times 2 \Rightarrow y \times 4 \Rightarrow 2^b = 4$	2
STEP 3 **Find b** **Why?** This line moves the solution one step closer to the answer. Copy the setup first, then simplify only one part at a time. $b = 2$	1
STEP 4 **Find a** **Why?** This line moves the solution one step closer to the answer. Copy the setup first, then simplify only one part at a time. $3 = a(1)^2 \Rightarrow a = 3$	1
STEP 5 **Result** **Why?** Write the cleaned-up result clearly and box it so the examiner can find it immediately. $\boxed{y = 3x^2}$	1
Total	7/7

FINAL ANSWER — WRITE THIS CLEARLY AND BOX IT

$$Y = \frac{3}{2}X$$

MODULE - 4

Probability Distributions

Every option: PMF/PDF plus Binomial, Poisson or Normal

Q7

20 Marks

7(a) For x=0,1,2,3 with P(X=x)=k,2k,3k,4k, find k, mean, variance and SD. **[06 Marks]**

Discrete PMF · 3/3 papers

FIRST UNDERSTAND THIS IN PLAIN ENGLISH

Probabilities must add to one. Then make $xP(x)$ and $x^2P(x)$ columns to get the mean and spread.

VTU ANSWER ORDER

1. Given / definition
2. Formula / theorem
3. Substitution
4. Stepwise working
5. Boxed answer

VTU-STYLE STEPWISE WORKING	MARKS
STEP 1 **Normalize** **Why?** All probabilities together must equal 1, so add or integrate them and solve for the missing constant. $10k = 1 \Rightarrow k = 0.1$	1
STEP 2 **Mean** **Why?** Multiply each value by its probability and add. This finds the balance point of the distribution. $E(X) = 0 + 0.2 + 0.6 + 1.2 = 2$	2

STEP 3

Second moment

Why? This line moves the solution one step closer to the answer. Copy the setup first, then simplify only one part at a time.

$$E(X^2) = 0 + 0.2 + 1.2 + 3.6 = 5$$

1

STEP 4

Variance

Why? Measure how far values spread from the mean. Variance is the squared spread; standard deviation is its square root.

$$\mathrm{Var}\,X = 5 - 2^2 = 1$$

1

STEP 5

SD

Why? This line moves the solution one step closer to the answer. Copy the setup first, then simplify only one part at a time.

$$\boxed{k = 0.1, \mu = 2, \sigma = 1}$$

1

Total 6/6

FINAL ANSWER — WRITE THIS CLEARLY AND BOX IT

2
0.1, 1
, ,
Σ
KM 2 Σ
== ==
1

7(b) Server requests per minute follow Poisson mean 2. Find P(0), P(2), and P(X≥3). [07 Marks]

Poisson distribution · 3/3 papers

FIRST UNDERSTAND THIS IN PLAIN ENGLISH

Poisson counts events in a fixed interval. For “at least 3,” subtract the easy cases 0, 1 and 2 from one.

VTU ANSWER ORDER

1. Given / definition	2. Formula / theorem
3. Substitution	4. Stepwise working
5. Boxed answer	

VTU-STYLE STEPWISE WORKING	MARKS
STEP 1 **Formula** **Why?** Write the standard rule first. It tells the examiner exactly which method you are using. $P(X = x) = e^{-2}2^x/x!$	1
STEP 2 **No request** **Why?** This line moves the solution one step closer to the answer. Copy the setup first, then simplify only one part at a time. $P(0) = e^{-2} = 0.1353$	2
STEP 3 **Exactly two** **Why?** This line moves the solution one step closer to the answer. Copy the setup first, then simplify only one part at a time. $P(2) = 2e^{-2} = 0.2707$	2

STEP 4

At least three

Why? This line moves the solution one step closer to the answer. Copy the setup first, then simplify only one part at a time.

$$\boxed{1 - e^{-2}(1 + 2 + 2) = 0.3233}$$

2

Total 7/7

FINAL ANSWER — WRITE THIS CLEARLY AND BOX IT

$P(0) = 0.1353$, $P(2) = 0.2707$, $P(X \geq 3) = 0.3233$

7(c) Scores of 1000 students are N(100,10²). Estimate counts above 120, below 90, and between 90 and 110. **[07 Marks]**

Normal distribution · 3/3 papers

FIRST UNDERSTAND THIS IN PLAIN ENGLISH

Convert every score to a z-score—how many standard deviations it is from the mean—then read the normal table and multiply by 1000.

VTU ANSWER ORDER

1. Given / definition
2. Formula / theorem
3. Substitution
4. Stepwise working
5. Boxed answer

VTU-STYLE STEPWISE WORKING	MARKS

STEP 1 **Standardize** **Why?** This line moves the solution one step closer to the answer. Copy the setup first, then simplify only one part at a time. $z = (x - 100)/10$	1
STEP 2 **Above 120** **Why?** This line moves the solution one step closer to the answer. Copy the setup first, then simplify only one part at a time. $1000[1 - \Phi(2)] = 22.8 \approx 23$	2
STEP 3 **Below 90** **Why?** This line moves the solution one step closer to the answer. Copy the setup first, then simplify only one part at a time. $1000\Phi(-1) = 158.7 \approx 159$	2
STEP 4 **Between 90 and 110** **Why?** This line moves the solution one step closer to the answer. Copy the setup first, then simplify only one part at a time. $\boxed{1000[2\Phi(1) - 1] = 682.6 \approx 683}$	2
Total	7/7

FINAL ANSWER — WRITE THIS CLEARLY AND BOX IT

23
,
159
,
683
STUDENTS

OR

Q8

20 Marks

8(a) Let f(x)=kx for 0<x<2 and 0 otherwise. Find k, E(X), and P(0.5<X<1.5). [06 Marks]

Continuous PDF · 3/3 papers

FIRST UNDERSTAND THIS IN PLAIN ENGLISH

The total area under a density must be one. Probabilities are areas between the requested limits.

VTU ANSWER ORDER

1. Given / definition
2. Formula / theorem
3. Substitution
4. Stepwise working
5. Boxed answer

VTU-STYLE STEPWISE WORKING	MARKS
STEP 1 **Normalize** **Why?** All probabilities together must equal 1, so add or integrate them and solve for the missing constant. $\int_0^2 kx\,dx = 1 \Rightarrow k = 1/2$	2

STEP 2 **Mean setup** **Why?** Multiply each value by its probability and add. This finds the balance point of the distribution. $$E(X) = \int_0^2 x(x/2)dx$$	1
STEP 3 **Mean** **Why?** Multiply each value by its probability and add. This finds the balance point of the distribution. $$E(X) = 4/3$$	1
STEP 4 **Probability setup** **Why?** This line moves the solution one step closer to the answer. Copy the setup first, then simplify only one part at a time. $$\int_{0.5}^{1.5} x/2\,dx$$	1
STEP 5 **Answer** **Why?** Write the cleaned-up result clearly and box it so the examiner can find it immediately. $$\boxed{k = 1/2, E(X) = 4/3, P = 1/2}$$	1
Total	6/6

FINAL ANSWER — WRITE THIS CLEARLY AND BOX IT

$1/2$
,
E $4/3$
(,
X
$K)$ P

$==$ $=$
$1/2$

8(b) If X~B(5,0.2), find P(X=0), P(X=2), and P(X≥1). **[07 Marks]**

Binomial distribution · 3/3 papers

FIRST UNDERSTAND THIS IN PLAIN ENGLISH

Binomial is for a fixed number of independent yes/no trials. “At least one” is easiest as one minus none.

VTU ANSWER ORDER

1. Given / definition
2. Formula / theorem
3. Substitution
4. Stepwise working
5. Boxed answer

VTU-STYLE STEPWISE WORKING	MARKS
STEP 1 **Identify values** **Why?** This line moves the solution one step closer to the answer. Copy the setup first, then simplify only one part at a time. $n = 5, p = 0.2, q = 0.8$	1

STEP 2 **None** **Why?** This line moves the solution one step closer to the answer. Copy the setup first, then simplify only one part at a time. $P(0) = q^5 = 0.32768$	2
STEP 3 **Exactly two** **Why?** This line moves the solution one step closer to the answer. Copy the setup first, then simplify only one part at a time. $P(2) = \binom{5}{2} p^2 q^3 = 0.2048$	2
STEP 4 **At least one** **Why?** This line moves the solution one step closer to the answer. Copy the setup first, then simplify only one part at a time. $\boxed{P(X \geq 1) = 1 - q^5 = 0.67232}$	2
Total	7/7

FINAL ANSWER — WRITE THIS CLEARLY AND BOX IT

0.32768
,
0.2048
,
0.67232

8(c) A bad reaction has probability 0.001. For 2000 people, find P(more than 2 reactions). [07 Marks]

Rare-event Poisson approximation · Exact pattern in 2/3 + model

FIRST UNDERSTAND THIS IN PLAIN ENGLISH

Many trials with a tiny probability behave like Poisson. The mean is np=2. Subtract cases 0,1,2 from one.

VTU ANSWER ORDER

1. Given / definition
2. Formula / theorem
3. Substitution
4. Stepwise working
5. Boxed answer

VTU-STYLE STEPWISE WORKING	MARKS
STEP 1 **Choose approximation** **Why?** This line moves the solution one step closer to the answer. Copy the setup first, then simplify only one part at a time. $\lambda = np = 2000(0.001) = 2$	1
STEP 2 **Poisson formula** **Why?** Write the standard rule first. It tells the examiner exactly which method you are using. $P(X = x) = e^{-2}2^x/x!$	1
STEP 3 **Complement** **Why?** This line moves the solution one step closer to the answer. Copy the setup first, then simplify only one part at a time. $P(X > 2) = 1 - [P(0) + P(1) + P(2)]$	2

STEP 4 **Substitute** **Why?** Put the known numbers or conditions into the formula carefully before simplifying. $= 1 - e^{-2}(1 + 2 + 2)$	2
STEP 5 **Answer** **Why?** Write the cleaned-up result clearly and box it so the examiner can find it immediately. $\boxed{P(X > 2) \approx 0.3233}$	1
Total	7/7

FINAL ANSWER — WRITE THIS CLEARLY AND BOX IT

$P(X > 2) \approx 0.3233$

MODULE - 5

Joint Distributions and Hypothesis Testing

3/3: joint PMF, z/t tests, Type errors and chi-square

Q9

20 Marks

9(a) Joint probabilities for X=0,1 and Y=1,2 are [[0.2,0.3],[0.2,0.3]]. Find E(X), E(Y), E(XY), covariance and correlation. **[06 Marks]**

Joint distribution · 3/3 papers

FIRST UNDERSTAND THIS IN PLAIN ENGLISH

Add rows and columns for marginal probabilities. Covariance checks whether X and Y move together beyond what their separate means predict.

VTU ANSWER ORDER

1. Given / definition	2. Formula / theorem
3. Substitution	4. Stepwise working
5. Boxed answer	

VTU-STYLE STEPWISE WORKING	MARKS
STEP 1 **Marginals** **Why?** Add across rows or down columns to remove the other variable and get one-variable probabilities. $P_X = (0.5, 0.5), \quad P_Y = (0.4, 0.6)$	1
STEP 2 **Means** **Why?** Multiply each value by its probability and add. This finds the balance point of the distribution. $E(X) = 0.5, \quad E(Y) = 1.6$	2

STEP 3

Product mean

Why? Multiply each value by its probability and add. This finds the balance point of the distribution.

$$E(XY) = 0.8$$

1

STEP 4

Covariance

Why? Measure how far values spread from the mean. Variance is the squared spread; standard deviation is its square root.

$$0.8 - (0.5)(1.6) = 0$$

1

STEP 5

Correlation

Why? Compare how the two variables move together. Positive means they usually rise together; negative means one falls as the other rises.

$$\boxed{\rho = 0}$$

1

Total 6/6

FINAL ANSWER — WRITE THIS CLEARLY AND BOX IT

$$E(X) = 0.5,\ E(Y) = 1.6,\ E(XY) = 0.8,\ \mathrm{COV} = 0,\ \rho = 0$$

9(b) n=100, $\bar{x}$=52, known σ=10. Test H_0:μ=50 against μ≠50 at 5%. **[07 Marks]**

Large-sample z-test · 3/3 papers

FIRST UNDERSTAND THIS IN PLAIN ENGLISH

See how many standard errors the sample mean is away from the claimed mean. Compare that z-distance with 1.96.

VTU ANSWER ORDER

1. Given / definition
2. Formula / theorem
3. Substitution
4. Stepwise working
5. Boxed answer

VTU-STYLE STEPWISE WORKING	MARKS
STEP 1 **Hypotheses** **Why?** Write the claim being tested as H_0 and the competing claim as H_1 before doing any calculation. $H_0 : \mu = 50,\ H_1 : \mu \neq 50$	1
STEP 2 **Critical rule** **Why?** Write the standard rule first. It tells the examiner exactly which method you are using. $\lvert z \rvert > 1.96 \Rightarrow \text{reject}$	1
STEP 3 **Standard error** **Why?** This is the usual amount a sample mean moves by chance. A smaller standard error means a more precise sample. $SE = 10/\sqrt{100} = 1$	1

STEP 4

Statistic

> **Why?** Turn the observed difference into standard-error units so it can be compared with the table value.

$$z = (52 - 50)/1 = 2$$

2

STEP 5

Decision

> **Why?** State the decision in words and connect it back to what the question actually asked.

$$2 > 1.96 \Rightarrow \text{reject } H_0$$

1

STEP 6

Conclusion

> **Why?** Multiply each value by its probability and add. This finds the balance point of the distribution.

The mean differs significantly from 50 at 5%.

1

Total 7/7

FINAL ANSWER — WRITE THIS CLEARLY AND BOX IT

$z = 2$; REJECT H_0

9(c) A die thrown 60 times gives frequencies 8,12,10,9,11,10. Test fairness at 5%; $\chi^2_{0.05,5}$=11.07. **[07 Marks]**

Chi-square goodness of fit · 3/3 papers

FIRST UNDERSTAND THIS IN PLAIN ENGLISH

A fair die expects 10 of each face. Measure each observed-versus-expected gap, square it, divide by expected, and add.

VTU ANSWER ORDER

1. Given / definition
2. Formula / theorem
3. Substitution
4. Stepwise working
5. Boxed answer

VTU-STYLE STEPWISE WORKING	MARKS
STEP 1 **Hypothesis and expected counts** **Why?** This line moves the solution one step closer to the answer. Copy the setup first, then simplify only one part at a time. $H_0 : \text{fair}, \quad E = 60/6 = 10$	2
STEP 2 **Formula** **Why?** Write the standard rule first. It tells the examiner exactly which method you are using. $\chi^2 = \sum (O - E)^2 / E$	1
STEP 3 **Contributions** **Why?** This line moves the solution one step closer to the answer. Copy the setup first, then simplify only one part at a time. $0.4 + 0.4 + 0 + 0.1 + 0.1 + 0 = 1.0$	2

STEP 4

Compare

Why? Compare the calculated test value with the allowed boundary. Crossing the boundary means the difference is statistically significant.

$$df = 5, \quad 1.0 < 11.07$$

1

STEP 5

Conclusion

Why? State the decision in words and connect it back to what the question actually asked.

There is insufficient evidence at 5% to conclude that the die is biased.

1

Total 7/7

FINAL ANSWER — WRITE THIS CLEARLY AND BOX IT

$X^2 = 1.0$; DO NOT REJECT H_0

OR

Q10

20 Marks

10(a) Explain sample, null hypothesis, Type I error and Type II error. [06 Marks]

Testing terminology · 3/3 supplied VTU papers; official Module 5 syllabus

FIRST UNDERSTAND THIS IN PLAIN ENGLISH

A sample is the small group we inspect. H_0 is the claim on trial. Type I is a false alarm; Type II is missing a real effect.

VTU ANSWER ORDER

1. Given / definition	2. Formula / theorem
3. Substitution	4. Stepwise working

5. Boxed answer

VTU-STYLE STEPWISE WORKING	MARKS
STEP 1 **Sample** **Why?** This line moves the solution one step closer to the answer. Copy the setup first, then simplify only one part at a time. A finite collection selected from a population for analysis.	1
STEP 2 **Null hypothesis** **Why?** This line moves the solution one step closer to the answer. Copy the setup first, then simplify only one part at a time. The default claim about a population parameter being tested.	1
STEP 3 **Type I error** **Why?** This line moves the solution one step closer to the answer. Copy the setup first, then simplify only one part at a time. Rejecting a true H_0; probability α.	1.5
STEP 4 **Type II error** **Why?** This line moves the solution one step closer to the answer. Copy the setup first, then simplify only one part at a time. Not rejecting a false H_0; probability β.	1.5

STEP 5

Decision interpretation

Why? State the decision in words and connect it back to what the question actually asked. | 1

$\boxed{\text{Type I=false alarm; Type II=miss}}$

Total | 6/6

FINAL ANSWER — WRITE THIS CLEARLY AND BOX IT

TYPE I: REJECT TRUE H_0; TYPE II: RETAIN FALSE H_0

10(b) Sample 8,9,10,11,12. Test $\mu=9$ at 5%; $t_{0.025,4}=2.776$. [07 Marks]

Student t-test · 3/3 papers

FIRST UNDERSTAND THIS IN PLAIN ENGLISH

The sample is small and population spread is unknown, so use t. Compare the mean gap with the estimated standard error.

VTU ANSWER ORDER

1. Given / definition
2. Formula / theorem
3. Substitution
4. Stepwise working
5. Boxed answer

VTU-STYLE STEPWISE WORKING	MARKS

STEP 1 **Hypotheses** **Why?** Write the claim being tested as H_0 and the competing claim as H_1 before doing any calculation. $H_0 : \mu = 9,\ H_1 : \mu \neq 9$	1
STEP 2 **Mean** **Why?** Multiply each value by its probability and add. This finds the balance point of the distribution. $\bar{x} = 10$	1
STEP 3 **Sample spread** **Why?** Measure how far values spread from the mean. Variance is the squared spread; standard deviation is its square root. $\sum (x - 10)^2 = 10,\ s^2 = 2.5,\ s = 1.5811$	2
STEP 4 **Statistic** **Why?** Turn the observed difference into standard-error units so it can be compared with the table value. $t = (10 - 9)/(1.5811/\sqrt{5}) = 1.414$	1

STEP 5

Compare

Why? Compare the calculated test value with the allowed boundary. Crossing the boundary means the difference is statistically significant.

$$1.414 < 2.776$$

1

STEP 6

Conclusion

Why? State the decision in words and connect it back to what the question actually asked.

$$\boxed{\text{Do not reject } H_0}$$

1

Total 7/7

FINAL ANSWER — WRITE THIS CLEARLY AND BOX IT

1.414
;
T
DO NOT REJECT
$=H$
0

10(c) Samples have $n_1=100$, $\bar{x}_1=50$, $s_1=8$ and $n_2=64$, $\bar{x}_2=47$, $s_2=6$. Test at 5% whether the population means are equal. [07 Marks]

Difference of means z-test · Tests for means are in the official syllabus and scanned papers

FIRST UNDERSTAND THIS IN PLAIN ENGLISH

Compare the gap between the two sample means with their combined standard error. A z-distance above 1.96 is significant.

VTU ANSWER ORDER

1. Given / definition	2. Formula / theorem
3. Substitution	4. Stepwise working

5. Boxed answer

VTU-STYLE STEPWISE WORKING	MARKS
STEP 1 **Hypotheses** **Why?** Write the claim being tested as H_0 and the competing claim as H_1 before doing any calculation. $H_0 : \mu_1 = \mu_2, \quad H_1 : \mu_1 \neq \mu_2$	1
STEP 2 **Combined standard error** **Why?** This is the usual amount a sample mean moves by chance. A smaller standard error means a more precise sample. $SE = \sqrt{\frac{8^2}{100} + \frac{6^2}{64}} = 1.0966$	2
STEP 3 **Test statistic** **Why?** Turn the observed difference into standard-error units so it can be compared with the table value. $z = \frac{50 - 47}{1.0966} = 2.735$	2
STEP 4 **Compare** **Why?** Compare the calculated test value with the allowed boundary. Crossing the boundary means the difference is statistically significant. $2.735 > 1.96 \Rightarrow \text{reject } H_0$	1

STEP 5 **Conclusion** **Why?** Multiply each value by its probability and add. This finds the balance point of the distribution. The population means differ significantly at the 5% level.	1
Total	7/7

FINAL ANSWER — WRITE THIS CLEARLY AND BOX IT

$Z = 2.735$; REJECT H_0